DATA ANALYSIS PROJECT REPORT

Bank Customer Churn Analysis

Identifying Customer Attrition Drivers and Retention Opportunities UsingSQL, Python,andPower BI

Timeline: July 2026

Sector: Banking / Financial Services

Executive Summary

This comprehensive analysis examines customer attrition within a retail banking environment using a dataset of 10,000 customers. With a baseline churn rate of 20.38%, the project leverages SQL and Python for deep-dive exploratory data analysis (EDA) and Power BI for executive visualization. Key findings highlight a critical link between customer complaints and churn (99.5% correlation) and significant geographical risk in the German market.

TOTAL LOST REVENUE (BALANCE)

185.68M

OVERALL CHURN RATE

20.38%

COMPLAINT CUSTOMER EXHIBIT

99.5%

HIGHEST CHURN RATE

32.44%

Business Problem & Objectives

Increased attrition is leading to direct revenue loss and diminishing customer lifetime value. This project aims to:

- Identify primary churn drivers and high-risk segments.
- Quantify financial impact to justify retention spend.
- Develop strategic recommendations for cross-selling and service improvement

Data Methodology

Tool	Purpose
SQL	Business Analysis & Complex Querying
Py thon	Exploratory Data Analysis (EDA) & Data Cleaning
Power BI	Dashboard Development & Visualization
Excel	Initial Validation

Key Findings & Insights

Analysis across various dimensions revealed the following critical insights:

Metric / Segment	Result / Finding
Highest Risk Region	**Germany (32.44% Churn)**
Age-Related Attrition	Customers 51+ years old show 44.65% churn.
Complaint Impact	99.5% of complaining customers churned.
Product Affinity	Lowest churn (7.6%) observed in 2-product segments.
Engagement Status	Inactive members churn at 26.87%.

Strategic Recommendations

- **Service Recovery:** Implement a 24-hour resolution protocol for all formal complaints.
- **Engagement Campaigns:** Target inactive members with personalized offers to re-establish banking habits.
- **Product Deepening:** Incentivize single-product customers to move to a 2-product bundle, as this carries the lowest risk profile.
- **Localization:** Audit German operations to understand the 32% churn rate vs. other regions.
- **High-Value VIP:** Assign relationship managers to high-balance customers to protect the €185M+ at-risk capital.

Future Roadmap

To move from descriptive to predictive analytics, the following phases are recommended:

1. Development of a **Machine Learning Churn Prediction Model**.
2. In-depth **Customer Lifetime Value (CLV)** analysis.
3. **Cohort Analysis** to track retention trends over time.
4. Integration of an **Early Warning System** into the CRM.